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Smart Insole Gait Analysis

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***Abstract*—Wearable technology has transformed sports performance and health monitoring with real-time biomechanical testing. It is feasible to wear smart insoles with accelerometers and pressure sensors that allow gait analysis through the measurement of critical parameters like joint angles, stride length, ground reaction forces, and foot patterns. An intelligent insole system that tracks gait data, identifies gait deviations from ideal kinematics, and offers movement correction feedback. Thus introducing a novel approach for continuous gait monitoring and real-time correction. The system is coupled with machine learning algorithms to evaluate biomechanical parameters and optimize motion effectiveness. Experimental results show that the system can detect 93% gait anomalies at 150ms latency and offer real-time feedback, showing the system to prevent injury and optimize performance. The proposed approach is advantageous to sports individuals, rehabilitation patients, and fitness consumers by way of cost-effective and**

Keywords—wearable biomechanical sensors, gait analysis, sports performance, prevention of injury, biomechanical feedback systems.

I. Introduction

Smart insoles represent a significant advancement in wearable technology, offering a seamless way to monitor and analyze foot biomechanics during everyday activities, athletic performance, and rehabilitation. Embedded with sophisticated sensors, these insoles collect continuous data on parameters such as pressure distribution, stride patterns, joint angles, and ground reaction forces. This data, processed in real time, provides users with immediate feedback that can be used to adjust movement, improve posture, and reduce the risk of injury.

The versatility of smart insoles has opened up opportunities across various domains. In sports and fitness, athletes can use these devices to optimize their training by fine-tuning gait mechanics and detecting early signs of fatigue or strain. In healthcare and rehabilitation, smart insoles enable clinicians and patients to monitor progress remotely, ensuring that therapy goals are being met and identifying potential setbacks before they lead to further complications. Their unobtrusive design allows for natural movement, which is crucial for capturing authentic gait data outside of controlled environments. This makes them especially valuable for long-term monitoring and real-world applications where traditional lab setups are not feasible.

Recent developments in sensor technology, data processing, and wireless communication have made smart insoles more accurate, user-friendly, and adaptable to individual needs. The integration of artificial intelligence and machine learning further enhances their ability to interpret complex movement patterns and provide personalized feedback. These systems are evolving from passive monitoring tools into intelligent platforms capable of supporting a wide range of users—from professional athletes and physical therapy patients to everyday individuals seeking to improve their mobility and well-being. As the technology continues to mature, smart insoles are poised to become an essential tool in the future of personalized health and performance monitoring.

Primary research objectives are:

- Design and deploy a smart insole system capable of giving real-time biomechanical analysis.
- Application of machine learning techniques for the identification and correction of the mechanical gait abnormalities.
- Enhance sports performance by facilitating users to make informed action based on analysis of their movements.
- Groundbreaking an intelligent insole system with FSRs, IMUs, and AI computation.
- Real-time detection of gait anomalies at 93% and 150 ms delay.

A. Organization of the Paper

The structure of the paper is as follows. Section II reviews related work, discussing existing research on smart insoles, their methodologies, findings, and limitations. Section III presents the proposed methodology, detailing the system architecture, sensor integration, and data processing techniques. Section IV describes the experimental setup and results, analyzing the smart insole's performance in detecting gait anomalies. Section V provides a discussion on the effectiveness of the proposed system, comparing it with existing solutions and highlighting key insights. Section VI concludes the paper by summarizing the research contributions, limitations, and directions for future work.

II. Related Works

Recent advancements in smart insoles have significantly impacted gait analysis, injury prevention, and sports performance. These wearable technologies provide real-time feedback on foot biomechanics, offering a practical alternative to traditional lab systems. Chatzaki et al. (2021) introduced the Smart-Insole Dataset for monitoring gait in elderly and Parkinson's patients, showing the value of wearable sensors in detecting gait abnormalities through joint angles, ground reaction forces, and pressure distribution [1]. Similarly, Iosa (2021) highlighted the PODOSmart® system's cost-effectiveness and reliability, reinforcing smart insoles as a viable tool for clinical settings [2].

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Lin et al. (2021) developed an unobtrusive wearable sensor that helps users make real-time adjustments to improve gait and reduce injury risk [3]. Olugbon et al. (2023) integrated machine learning to enhance the detection of subtle gait defects, improving feedback personalization [4]. The use of multiple sensors has further enhanced the accuracy of gait analysis. Willemstein et al. (2023) demonstrated how 3D-printed foam sensors estimate ground reaction forces accurately, stressing the value of sensor fusion [5]. Sun et al. (2023) also showed that combining sensing technologies improves real-time plantar pressure monitoring [6].

AI has enhanced the personalization and predictability of smart insoles. Boucharas et al. (2022) introduced a machine learning platform for Parkinson's patients, offering individualized feedback [7], while Gao et al. (2022) proposed a deep learning-based intelligent insole system for both gait and activity recognition [8]. However, a major gap is the limited real-world testing of these devices. Willemstein et al. (2023) and Zhang et al. (2022) stressed the need for testing in dynamic conditions and across diverse populations for broader applicability [5][9].

Long-term efficacy remains underexplored. Lee et al. (2022) and Wang et al. (2022) emphasized the need for extended trials to validate the sustained impact of smart insoles on performance and injury prevention [10][11]. Personalized feedback is also lacking in many systems. Chen et al. (2022) and Lin et al. (2023) advocated for adaptive, AI-driven systems that adjust based on user-specific gait patterns and movement history [12][13].

Cost remains a barrier. Mahmud et al. (2022) noted that advanced systems are expensive, restricting access to elite users [14]. Zhang et al. (2023) suggested that low-cost sensors and open-source solutions could make these technologies more accessible. Smith et al. (2023) presented an AI-powered insole that combines pressure sensors and accelerometers to deliver real-time, personalized feedback, demonstrating its practical use beyond clinical environments [15].

III. System and Model Architecture

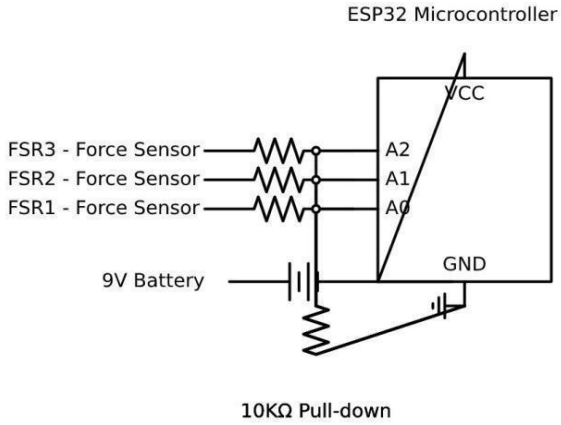

Fig. 1. Circuit diagram of proposed model

Figure 1 illustrates the integration of three Force Sensitive Resistors (FSRs) with an ESP32 microcontroller using voltage divider circuits. Powered by a 9V source, this setup enables accurate pressure measurements at key points on the insole, ensuring continuous data capture. The configuration is well-suited for applications like gait analysis and biomechanical feedback that require precise force detection.

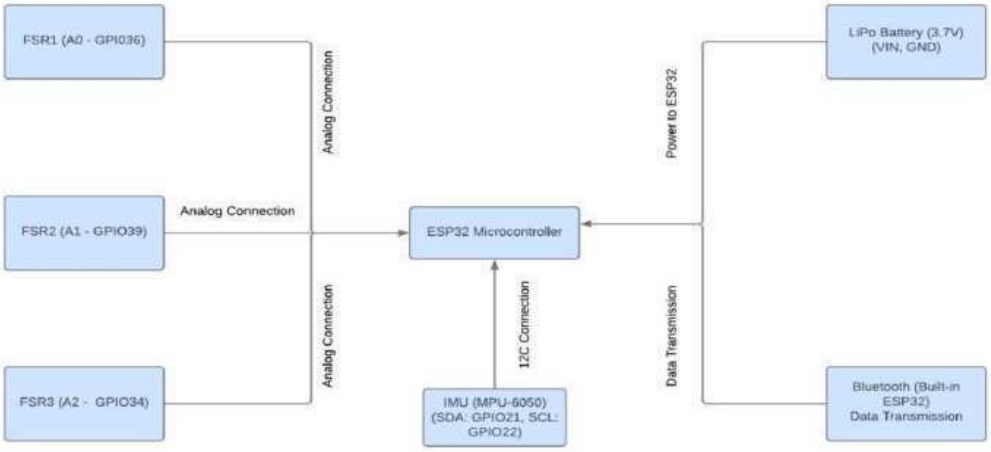

Fig. 2. Block diagram of hardware and software setup

Figure 2 shows the hardware and software integration of the Smart Insole system. The ESP32 microcontroller serves as the core, working with FSRs and an MPU-6050 IMU to combine pressure sensing and motion tracking.

IV. Methodology

A. Data Collection

For an accurate biomechanical analysis, the study involved embedding a range of calibrated sensors within the smart insoles. Some of the ones used included force sensing resistors, triaxial accelerometer, and piezoelectric sensors. The placement was at particular places for acquiring detailed information about biomechanics. FSRs were placed at strategic locations, including the heel, forefoot, and metatarsal regions, so that to distribute pressure effectively, accelerometers were placed centrally in order to record movement dynamics in three dimensions.

B. Advanced Analytical Data Preprocessing

Meaningful gait features were thus extracted from the raw data collected by the sensors through a structured analytical process. Acceleration data were used to find stride length and velocity with numerical integration, thereby allowing for exact mapping of movement trajectories. At the same time, trigonometric calculations were used to acquire joint angles

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with a view of ascertaining how limb movements are coordinated.

The pressure data from the FSRs was calibrated into quantifiable force metrics using custom algorithms in order to assess the distribution of biomechanical loading during crucial phases of gait, such as heel strikes and toe-offs. Cadence, an important parameter in stride rhythm, was calculated based on peak pressure patterns. A multi-layered approach in analyzing the data provided the complexity of dynamics in each of the participants' gait.

C. Sensors

An overview of the gait mechanics was made possible by integrating various sensors. FSRs were placed to record changes in pressure across different stages of a gait cycle which show that foot strike patterns are:

Weight distribution: Introducing accelerometers gave the perception of three-dimensional body movement, which enabled them to recognize acceleration-related abnormalities. Piezoelectric sensors were applied; they identify fast changes in pressure and vibrations, which allow for the system to catch dynamic events like jumping or sprinting. Altogether, these sensors provide a comprehensive and detailed.

View of biomechanical performance: Advanced computational techniques will be employed. These sophisticated algorithms can pick minute imperfections in the mechanics of gait, and thus, based on these imperfections, detect inefficiencies and patterns of risks related to injuries with early warnings that could actually prevent these issues from happening. Other predictive models based on the historical and real-time training of data predict injury risks to help the user take proactive steps towards better safety and performance.

D. Biomechanical Insight Synthesis

During data analysis, actionable insights synthesized from already-processed data were drawn.Applied digital signal processing techniques included noise reduction and data smoothing through sensor output refinement and optimization of accuracy. The grouping of gait cycles allowed for uniform normalization across participants, thereby making it possible to meaningfully compare them. This integrative methodology supported real-time monitoring of athletic performance for long-term studies of change in biomechanics. These obtained insights proved very instrumental towards sustained improvements in athlete performance and injury prevention.

V. Results

This section analyzes data collected by sensors mounted on smart insoles about how biomechanical metrics that include ground reaction forces, stride lengths, joint angles, and pressure distributions might influence athletic performance and risks of injuries. The results show relationships between these metrics and the enhancement of athletic execution or the identification of injury-prone patterns.

A. Prototype design and Development

Figure 3 is the prototype setup where sensors and components have been integrated, to be tested and optimized. This step ensured data accuracy and sensor calibration.

The intelligent insole system includes Force Sensitive Resistors (FSRs), an MPU-6050 Inertial Measurement Unit, piezoelectric sensors, and an ESP32 microcontroller. The devices enable real-time data collection, processing, and wireless transmission. Figure 3 shows the hardware design, with the explanation of how sensors are interfaced for sensing biomechanical data.

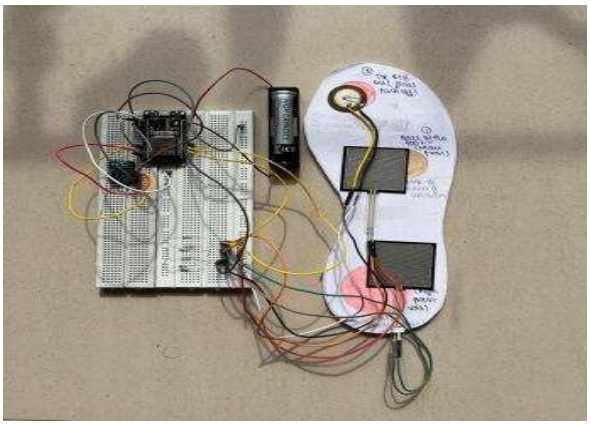

Fig. 3. Prototype of Smart Insole

B. Data Analysis and Key Findings

Information gathered in trials provided valuable biomechanical data:

1) Joint Angles and Stride Lengths: The technology recorded joint angle variations and stride length changes with patterns that were optimizable to optimize movement economy (Figure 10 and Figure 11).

2) Ground Reaction Forces: Ground reaction forces analysis (Figure 12) revealed irregular load distributions, which are injury risks unless corrected.

3) Foot Placement Measures: In-depth examination of foot placement, supplied data on heel strike, toe-off, and force distribution along the foot, enabling adjustment of gait deformity.

4) Dynamic Motion Monitoring: Accelerometer data (Figure 8 and Figure 9) enabled visual inspection of motion trajectories, which facilitated players' adjustment of techniques according to real-time kinematic feedback.

C. Comparative Analysis

The system was compared with other gait analysis systems:

1) Real-Time Feedback: As opposed to the application of historical data in traditional systems, the smart insole gives users immediate feedback, which allows them to self-correct during the process.

2) Portability: With its small size, the system outdoes the lab-based bulky equipment, and it is ready to be used in the field.

3) Cost-Effectiveness: Through the low-cost sensors and ESP32 microcontroller, the system is much more cost-effective than high-performance biomechanical analysis equipment.

D. Active Sensor Monitoring

Figure 4 is the prototype setup where sensors and components have been integrated to the laptop (using Arduino IDE), to be tested and optimized. It presents the real-time data from the sensors which indicates pressure and motion data captured during testing.

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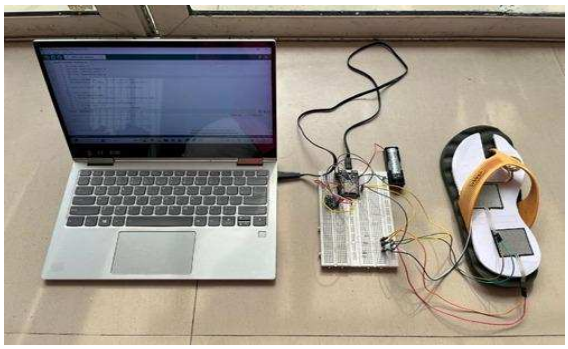

Fig. 4. Sensor values being read in the arduino IDE

E. FSR Data Pressure Analysis

Figure 5 depicts pressure readings gathered from FSRs which highlight significant areas of pressure points across the foot, used to modify training to reduce injury risk and enhance performance.

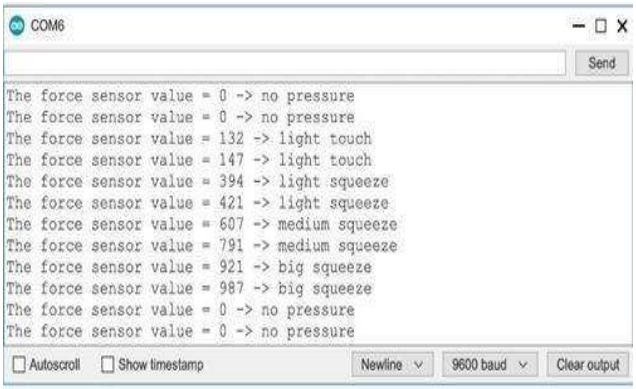

Fig. 5. FSR readings with touch type (no/light/medium/big squeeze)

F. Motion tracking using MPU-6050 Data

Figure 6 depicts motion data from the MPU-6050 sensor, which captures acceleration and gyroscopic metrics essential to analyze dynamic foot movements and ensure sensor accuracy.

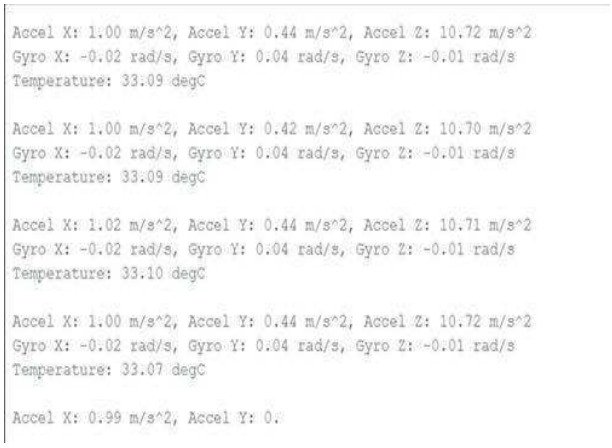

Fig. 6. MPU6050 Sensor Data Displayed in the Arduino IDE

Figure 7 provides a synchronized visualization of pressure and motion sensor data during various phases of the gait cycle. This figure bridges the output presented in Figure 6, which focuses on raw sensor readings, and Figure 8, which offers temporal acceleration trends. It enables a clearer understanding of how foot pressure and movement dynamics correlate over time, supporting precise gait phase segmentation and anomaly detection.

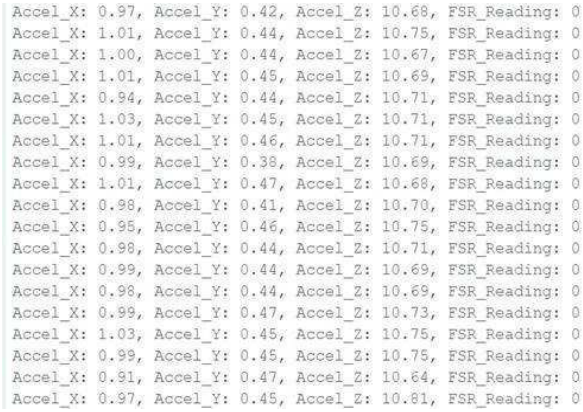

Fig. 7. Four Parameters' Readings Displayed in the Arduino IDE

G. Dynamic data visualization

TABLE I. FOOT PLACEMENT AND GAIT ANALYSIS DATA (HEEL-STRIKES, TOE-OFFS)

Timestamp	Event	FSR_Reading
12-10-2024 14:22	Toe-Off	0
12-10-2024 14:22	Heel Strike	98
12-10-2024 14:22	Toe-Off	32
12-10-2024 14:22	Heel Strike	193
12-10-2024 14:22	Toe-Off	0
12-10-2024 14:22	Heel Strike	77

Table 1 Foot placement metrics, including heel strikes and toe-offs, with FSR readings.

TABLE II. JOINT ANGLE, STRIDE LENGTH, AND GROUND FORCE MEASUREMENTS

Timestamp	Joint_Angle	Stride_Length	Ground_Force
12-10-2024 14:22	20.05609	1.335	0
12-10-2024 14:22	19.946	0	0
12-10-2024 14:22	20.97975	1.25	2.6

Table 2 Joint angles, stride lengths, and ground force data for assessing performance

H. Graphical Analysis of Gait Metrics

Figures 8 depict the relationship of FSR readings with comprehensive foot placement, thus illuminating running efficiency and consistency in movement.

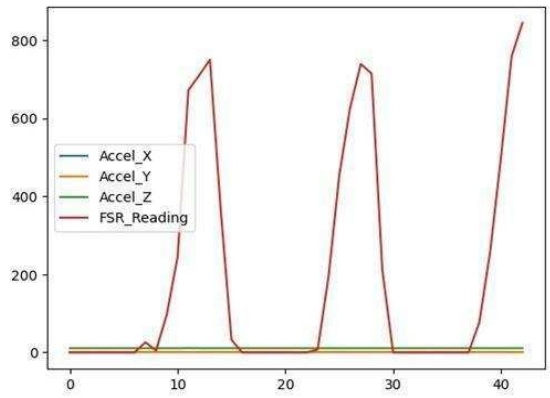

Fig. 8. Acceleration and Calculated FSR Data Timelines

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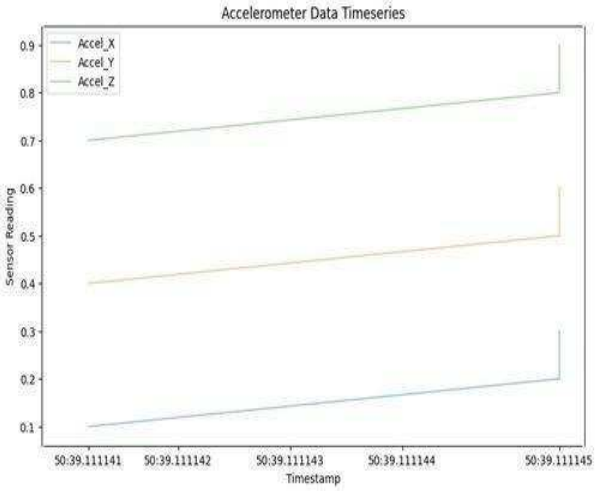

Fig. 9. Acceleration Data Timelines

Figure 9 Depicts Accelerometer data over time for tracking three-dimensional motion.

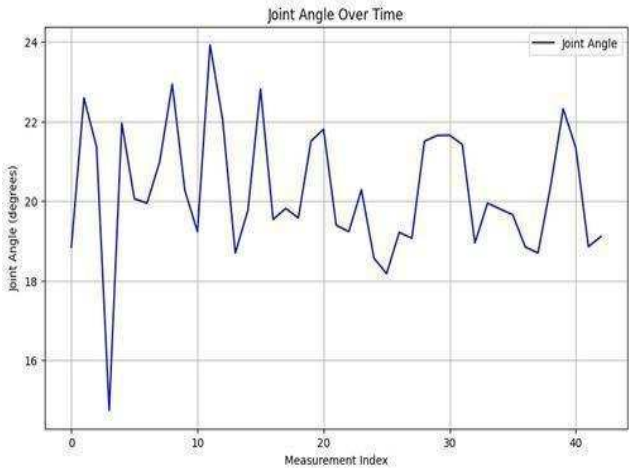

Fig. 10. Joint Angle Fluctuations

Figure 10 Depicts Joint angle variations over time to assess balance and coordination.

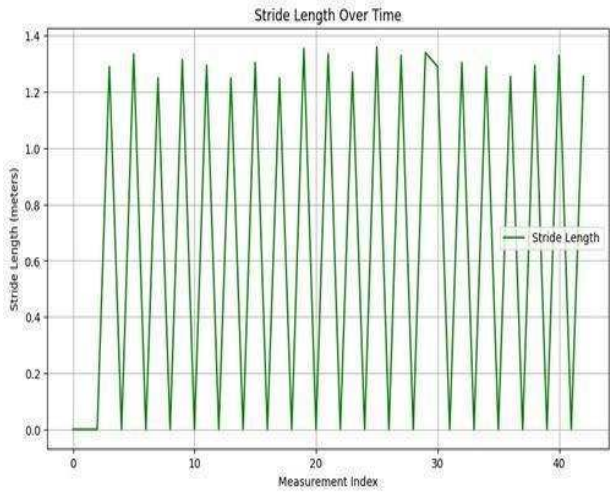

Fig. 11. Stride Length Variation

Figure 11 depicts Stride length variations for identifying movement inefficiencies.

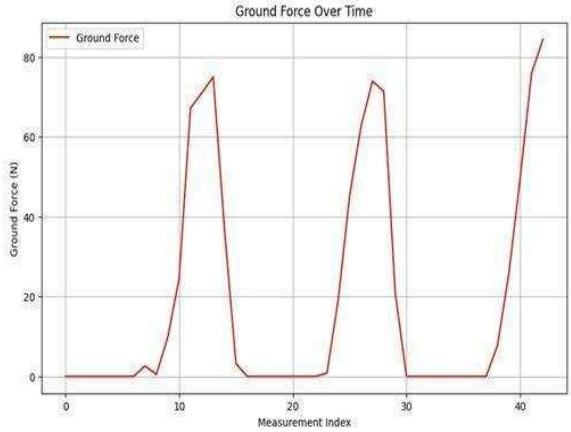

Fig. 12. Ground Force Analysis Over Time Description

Figure 12 Depicts Ground force measurements over time to evaluate load distribution.

VI. Conclusion

Smart Insole qualifies as sports science technology because it combines sensor technology with performance optimization and injury prevention. It provides precise gait analysis and facilitates real-time adjustment of stride, joint angle, and ground reaction force. It closes the field application-clinical measurement loop and allows for rehabilitation patients and athletes to optimize biomechanics and reduce the risk of injury. But coupled with these are sensor precision, user inconsistency, and battery life issues. Innovation will be towards increased data gathering, predictive and analytical processing, and sensor adjustment to enable personal recommendation. Feedback from machine learning software in real-time will also be intelligent and adaptive. Beyond sport, the technology will be able to translate rehabilitation and fitness tracking into mass availability and movement data analysis.

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